

FEW EXAMPLES OF ONLINE THREATS

Threat Type	Description	Basic Countermeasures	Examples
Malware & Viruses	Software designed to damage systems, steal data, or create backdoors.	Install reputable antivirus, keep systems updated, avoid unknown downloads.	Trojan, Spyware, Worm
Phishing	Fraudulent communication appearing from reputable sources to steal sensitive information.	Verify the sender's credibility, avoid clicking on suspicious links or attachments.	Email scams, Fake website forms
Ransomware	Malware that encrypts the victim's files, demanding ransom for decryption keys.	Regularly backup data, use security software, don't pay the ransom.	WannaCry, Petya
Social Engineering	Techniques used to manipulate people into providing confidential information or accessing secure systems.	Educate on common tactics, promote verification processes, enforce strict security policies.	Pretexting, Baiting, Scareware
DDoS Attacks	Overwhelming a system's resources to render it inaccessible.	Implement security measures to mitigate traffic, use DDoS protection services.	High traffic to websites, server overload
SQL Injection	Inserting malicious code into SQL queries to manipulate databases.	Use input validation, employ parameterized queries, and regularly update applications.	Unauthorised data access, data theft
Zero-day Exploits	Attacks exploiting vulnerabilities in software before the developer has issued a patch.	Keep software up to date, use intrusion detection systems, sandboxing environments.	Software bugs, Unpatched software

- **Appear to be from legitimate sources:** Banks, social media platforms, or even trusted colleagues.
- **Create a sense of urgency:** Threatening account suspension, requesting immediate action, or highlighting a "limited-time offer."
- **Contain links that lead to fake websites:** Designed to steal login credentials, financial information, or infect the device with malware.

In a similar way, social engineering tries to exploit human psychology to manipulate individuals into compromising security measures. Common tactics include:

- **Pretexting:** Creating a false scenario to gain trust and access to personal information.
- **Baiting:** Luring users with irresistible offers or exploiting their curiosity to click malicious links or attachments.
- **Quid pro quo:** Offering seemingly helpful solutions in exchange for sensitive information.

Combating Phishing and Social Engineering isn't always easy, but you can learn to stay on your guard by doing the following:

- **Be cautious of unsolicited emails or messages:** Do not click on links or attachments from unknown senders.
- **Verify the sender's identity:** Contact the supposed sender through a trusted channel to confirm the legitimacy of the communication.
- **Be wary of urgency or emotional manipulation:** Do not rush into any actions based on pressure tactics.
- **Value your data:** Never share personal information through email or unverified websites.

Ransomware

The term ransomware has become extremely popular over the past decade, and for good reason. Ransomware is a particularly disruptive form of malware that encrypts a user's files, rendering them inaccessible. The attacker then demands a ransom payment typically in cryptocurrency for the decryption key. In terms of impacts of ransomware, victims can lose access to critical data like photos, documents, and business files. Businesses face operational disruptions and potential financial losses.

Preventing Ransomware Attacks:

- **Regular backups:** Ensure you have a recent, uninfected copy of your important data stored securely offline.
- **Software updates:** Keep your operating system, applications, and security software up-to-date to address vulnerabilities.

